

WLODAWA, Poland

WMO: 124970

Lat: 51.5533N

Lon: 23.5294E

Elev: 179

StdP: 99.19

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-13.9	-20.1	0.6	-17.1	-16.7	0.9	-13.3	11.5	2.8	10.5	2.3	2.6	10	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.3	20.4	28.4	19.5	26.6	18.7	21.5	28.0	20.5	26.4	19.6	24.7	4.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.3	23.9	18.5	13.6	22.9	17.7	13.0	22.1	63.2	28.2	59.9	26.7	56.7	24.7	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	DB	-20.5	33.1	4.4	1.4	-23.7	34.1	-26.2	34.9	-28.7	35.7	-31.9	36.8	(4)
(5)			WB	-20.8	23.2	4.3	0.9	-23.9	23.9	-26.4	24.4	-28.8	24.9	-32.0	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-3.1	-1.6	2.4	8.7	13.7	16.9	19.3	18.5	13.5	8.3	3.5	-1.2
	DBStd	8.92	5.70	5.40	4.58	4.24	3.86	3.40	3.19	3.17	3.63	4.16	4.48	5.18
	HDD10.0	1689	405	324	237	73	10	0	0	0	9	84	199	347
	HDD18.3	3797	663	557	494	289	151	67	27	36	149	312	446	605
	CDD10.0	1070	0	0	2	34	125	208	288	264	116	31	3	0
	CDD18.3	136	0	0	0	1	7	25	57	42	5	0	0	0
	CDH23.3	1381	0	0	0	12	87	251	546	427	58	0	0	0
	CDH26.7	372	0	0	0	2	9	53	167	131	9	0	0	0
Wind	WSAvg	3.7	4.3	4.2	4.3	3.7	3.3	3.2	3.1	3.0	3.2	3.8	4.1	4.4
Precipitation	PrecAvg	567	31	28	32	40	64	66	83	64	52	39	36	32
	PrecMax	749	71	56	75	90	151	200	219	272	114	97	64	73
	PrecMin	398	4	2	6	13	22	6	12	4	9	5	3	12
	PrecStd	82	18	13	16	18	29	36	48	49	32	27	15	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	10.8	17.0	24.8	28.0	30.6	32.8	32.7	27.9	21.5	15.6	9.9
		MCWB	6.6	7.7	10.5	15.2	18.5	21.0	20.7	20.4	18.3	16.0	11.9	8.6
	2%	DB	6.4	8.5	13.6	21.4	25.8	28.4	30.6	30.0	24.9	19.0	13.0	7.7
		MCWB	5.0	6.5	8.4	13.5	17.5	19.6	20.4	20.3	17.3	14.2	10.8	6.6
	5%	DB	4.9	6.7	11.3	18.6	23.8	26.2	28.5	27.8	22.7	16.7	11.2	6.2
		MCWB	3.7	5.1	7.4	11.7	16.4	18.6	19.8	19.4	16.3	13.5	9.6	5.2
	10%	DB	3.5	5.0	9.1	16.5	21.6	24.4	26.4	25.5	20.3	14.6	9.5	4.8
		MCWB	2.6	3.7	6.2	10.7	15.1	17.7	19.0	18.2	15.2	12.1	8.3	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.2	11.1	15.7	19.9	21.9	22.7	22.5	19.4	16.6	12.6	8.7
		MCDB	8.0	10.0	16.1	23.9	26.0	28.8	29.6	29.6	25.7	20.2	14.6	9.8
	2%	WB	5.2	6.7	9.2	14.2	18.2	20.7	21.6	21.2	18.1	15.2	11.1	6.8
		MCDB	6.2	8.3	12.9	20.1	23.9	26.8	28.1	27.8	23.5	18.1	12.8	7.6
	5%	WB	3.9	5.2	7.7	12.7	17.1	19.5	20.7	20.1	17.0	13.8	9.7	5.3
		MCDB	4.7	6.6	10.4	17.4	22.6	24.5	26.4	25.9	21.3	16.2	11.0	6.0
	10%	WB	2.7	3.7	6.5	11.3	15.8	18.4	19.8	19.0	15.8	12.3	8.3	4.0
		MCDB	3.5	4.9	8.8	15.2	20.6	22.8	24.7	23.7	19.4	14.3	9.3	4.7
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	7.3	9.9	10.6	10.5	10.7	11.1	9.7	8.0	5.1	4.4
		MCDBR	4.7	7.0	11.3	14.0	13.9	13.4	14.1	14.6	13.6	11.2	8.0	5.2
		MCWBR	4.1	5.2	7.1	7.2	6.7	6.1	5.5	6.0	6.7	6.9	5.7	4.4
	5% WB	MCDBR	4.5	6.7	10.1	12.7	12.8	12.1	12.0	12.5	11.9	10.3	7.2	5.1
		MCWBR	4.0	5.3	6.7	7.0	6.5	6.2	5.3	5.7	6.4	6.7	5.4	4.5
Clear-Sky Solar Irradiance	taub		0.312	0.340	0.376	0.427	0.415	0.411	0.438	0.434	0.397	0.363	0.349	0.309
	taud		2.234	2.207	2.214	2.146	2.246	2.282	2.219	2.226	2.292	2.363	2.296	2.276
	Ebn at Noon		678	763	809	807	840	845	813	793	780	736	628	619
	Edh at Noon		69	94	113	137	130	126	133	125	105	81	67	57
All-Sky Solar Radiation	RadAvg		0.76	1.50	2.75	4.11	5.19	5.55	5.24	4.66	3.22	1.86	0.79	0.53
	RadStd		0.13	0.21	0.38	0.49	0.52	0.45	0.54	0.36	0.50	0.30	0.08	0.10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	+74	N/A
(47) Regional (16 neighbors)		+0.55	N/A	N/A	+0.73	+0.45	N/A	N/A	-180	+90	+37

Nomenclature: See separate page